



UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment 1 CLO Mapping: CLO1

Course Code	MAT233	Semester	Spring 2026
Course Title	Linear Algebra		
Instructor	Adnan Rafiq Siddiqui		
Assignment Given Date	10-03-2026	Submission Date	19-03-2026
Total Marks	25		

Sr.	Name of student	Registration no.	Section	Assignment marks	Total marks
1					

Question No.1 [Marks: 10]

Using GeoGebra, **Describe** and visualize the three possible types of solutions for a system of linear equations in three variables: **unique solution**, **infinite solutions**, and **no solution**. For each case, provide a supporting example of a system of equations and explain the geometric interpretation (point, line, or parallel planes).

Question No. 2 [Marks: 7]

Find the values of a for which the system has no solutions, exactly one solution, or infinitely many solutions

$$\begin{aligned}x + y + 7z &= -7 \\2x + 3y + 17z &= -16 \\x + 2y + (a^2 + 1)z &= 3a\end{aligned}$$

Assume any 3 appropriate values of a to plot, using GeoGebra, corresponding planes which represent no solution, exactly one solution and infinitely many solutions. Attach the plots with your work.

Question No.3 [Marks: 8]

An **investment portfolio manager** is analyzing the performance of four financial instruments held by a client. Each instrument contributes differently to the portfolio's overall return, risk, liquidity, and tax exposure. The four instruments are tracked using the following variables:

- x : units held in **equity stocks**
- y : units held in **government bonds**
- z : units held in **commodity futures**
- w : units held in **real estate investment trusts (REITs)**

Based on four different financial audits and regulatory compliance checks, the portfolio manager derives the following four constraint equations. Note that **negative coefficients** reflect instruments that *reduce* a particular metric (e.g., stocks reducing net tax liability, or futures reducing liquidity scores):

Audit 1 (Return Balance): $2x - y + 3z - w = -1.5$

Audit 2 (Risk Assessment): $-x + 4y - 2z + w = 11$

Audit 3 (Liquidity Check): $3x + 2y - z - 2w = 2$

Audit 4 (Tax Exposure): $-2x - y + 4z + 3w = 7$

Each equation represents a financial constraint derived from a separate audit report. Your task is to determine whether this system has a **unique**, **infinite**, or **no solution**, and interpret the result in terms of portfolio management — specifically, whether a consistent and precise allocation of units across all four instruments exists that simultaneously satisfies every audit requirement.

Solve the system of linear equations using the **Gauss-Jordan elimination method**.

Express all solutions to **one decimal place** where applicable.